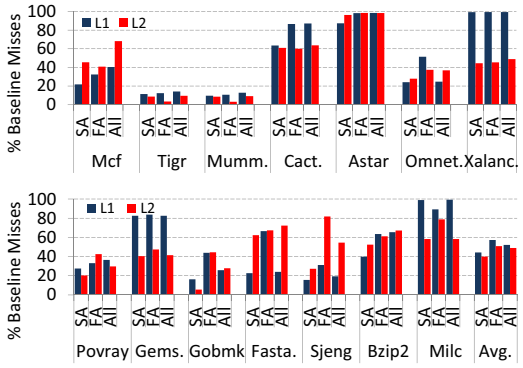


Figure 18: Percentage of baseline TLB misses eliminated using CoLT-SA, CoLT-FA, and CoLT-All.

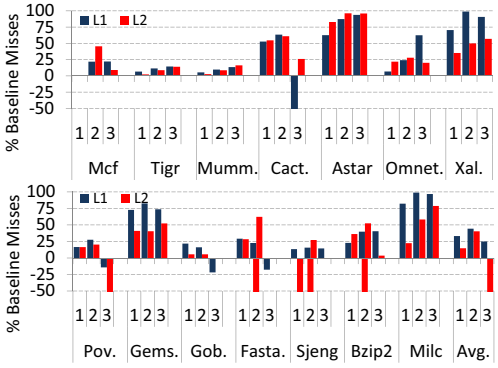


Figure 19: Percentage of baseline TLB misses eliminated by CoLT-SA when left-shifting the index by 1, 2, and 3 bits.

Figure 19 clearly shows that left-shifting the index bits by two provides the best balance between coalescing opportunity and conflict misses. Below this (left-shifting by one bit), we can only coalesce two entries, restricting our TLB miss elimination rates. However, left-shifting by three bits actually *increases* TLB misses in many cases due to the additional conflict misses. In general, unless there is very high contiguity, like for `Mummer`, `Tigr`, and `Milc`, left-shifting the index bits by three is overly-aggressive. Henceforth, we assume a left-shit of two bits for our indexing scheme.

7.1.3. Impact of bringing missing entries into L2 TLB for CoLT-FA and CoLT-All. As previously detailed, while CoLT-FA and CoLT-All bring coalesced entries into the fully-associative, superpage TLB, they also leverage the L2 TLB. For CoLT-FA, when a coalesced entry is brought into the superpage TLB, just the requested entry is also brought into the L2 TLB; for CoLT-All, a coalesced entry (where the coalescing amount is restricted by the index scheme) is brought into the L2 TLB. As we previously noted, this is useful since the superpage TLB is small (8-entry); as a result, only entries with high levels of coalescing are maintained there. As such, intermediate-level coalesced entries are often evicted. Bringing these entries into the L2 TLB as well increases the chance that these they remain available if necessary.

For CoLT-FA, we have run experiments to compare the case when (1) a coalesced entry is brought into the superpage TLB and just the translation triggering the coalescing is also brought into the L2 TLB, and (2) a coalesced entry is brought into the superpage TLB but the L2 TLB remains unaffected. We have found that on average, (1) outperforms (2) by an additional miss elimination of 10-15% for both L1 and L2 miss counts. We see particularly high gains with our approach on workloads with relatively lower contiguity such as `Povray`, since the small superpage TLB cannot coalesce a high enough number of entries to prevent eviction.

For CoLT-All, we have similarly run experiments to compare the case when (1) a coalesced entry is brought into the superpage TLB and its smaller coalesced version (a maximum of coalescing of four translations in our design) is brought into the L2 TLB, and (2) a coalesced entry is brought into the superpage TLB but the L2 TLB remains unaffected. We see again that our approach, (1) outperforms (2) by an average of 10-20% TLB miss eliminations.

7.1.4. Studying CoLT's effectiveness at higher associativities. We now consider CoLT effectiveness as TLB associativity is varied. A number of past studies have quantified how effectively increasing TLB associativity eliminates misses [13]. Generally, these studies have concluded that the slightly higher TLB hit rates are offset by huge power dissipation problems [7]. These observations are largely responsible for the relatively low associativity (typically 2-way or 4-way) supported on current TLBs.

CoLT, however, increases the benefits of higher set-associativity since the indexing scheme of the TLB can be more aggressively changed without as significant an increase in conflict misses. Overall, this allows higher levels of coalescing. Figure 20 compares how many L2 TLB misses in a 4-way 128-entry L2 TLB can be eliminated by CoLT-SA (4-way, CoLT-SA), by varying the associativity to 8-way but not allowing coalescing (8-way, No CoLT), and by allowing CoLT on the 8-way TLB (8-way, CoLT-SA). Note that all configurations use a fixed TLB size despite associativity changes.

First, Figure 20 shows that merely increasing the associativity to 8-way only eliminates 10% of the baseline L2 misses. In fact, even 4-way L2 TLBs with low-overhead CoLT-SA far exceed the benefits of higher associativity, eliminating 40% of baseline misses on average.

Figure 20 shows however, that the 8-way configuration augmented with CoLT-SA does provide significant benefits. CoLT-SA now eliminates 60% of the baseline misses on average, a substantial improvement over the other two scenarios. While a detailed power analysis is beyond the scope of this work, the performance, power ratio may therefore become more amenable with CoLT.

7.2. Performance Analysis

Up to this point, we have evaluated the benefits of CoLT in terms of miss rate eliminations. While this does indicate CoLT's effectiveness, we now focus on performance numbers which track how much faster each application runs with coalescing. Figure 21 details, for every benchmark, performance improvements from CoLT-SA, CoLT-FA, and CoLT-All. It also provides data on performance improvements that would occur with absolutely perfect, 100%-hit rate TLBs. The latter serves as a comparison point to determine how effectively CoLT performs. Once again, the baseline is a system with 4-way 32-entry and 128-entry L1 and L2 TLBs, and a 16-entry superpage TLB. CoLT-FA and CoLT-All conservatively reduce the superpage TLBs to 8 entries. Moreover, as previously detailed, we simulate a 4-way out-of-order processor.

Figure 21 shows that perfect TLBs would improve most benchmark runtimes by over 10% (eg. all except `Gobmk` and `Sjeng` in our benchmarks). In fact, `Xalancbmk` sees a huge 115% improvement in performance from TLBs that achieve 100% hit rate. These numbers indicate that TLB miss handling does significantly slow down bench-